WEEK 4 ASSESSMENT

Data + ML Fundamentals Review

Quiz • Preprocessing Task • Concept Viva

Date	28-Oct-2026	**Duration**	4 hours
Difficulty	Intermediate	**Total marks**	100

Assessment Instructions

- Complete independently. Use notes only if the trainer permits them.
- Write short explanations where requested and do not copy code without understanding it.
- You may be asked to explain any line of your notebook during the viva.
- Submit quiz answers and your preprocessing notebook.
- Suggested timing: Quiz 45 minutes; practical task 2 hours; debugging, reflection and viva 1 hour 15 minutes.

Part A — Multiple-Choice Quiz (20 marks)

Choose one best answer for each question. Each question carries 1 mark.

Which statement best describes Machine Learning?

- **A. A system learns patterns from examples**
- B. A program only follows fixed rules
- C. A database stores files
- D. A network connects devices

Predicting whether a transaction is fraudulent or legitimate is usually:

- A. Regression
- **B. Classification**
- C. Clustering
- D. Data compression

Predicting a numeric house selling price is usually:

- A. Classification
- B. Clustering
- **C. Regression**

- D. One-hot encoding

Grouping customers by behaviour without existing group labels is:

- A. Supervised regression
- B. Supervised classification
- **C. Unsupervised clustering**
- D. Scaling

In a dataset used to predict churn, the column churn is the:

- A. Scaler
- **B. Label / target**
- C. Test set
- D. Imputer

Which is a possible feature for predicting house price?

- A. The price to be predicted
- **B. Number of bedrooms**
- C. The model's final score
- D. The prediction itself

Why keep a test set separate?

- A. To enlarge the dataset
- **B. To estimate performance on unseen data**
- C. To fill missing values
- D. To encode categories

A missing-value imputer is used for:

- **A. Handling missing entries**
- B. Splitting rows
- C. Choosing the target automatically
- D. Measuring accuracy

A common strategy for a missing numeric feature is:

- A. Replace values with the column name
- **B. Use the median learned from training data**

- C. Copy values from the target
- D. Convert numbers to city names

StandardScaler generally:

- A. Converts categories to text
- **B. Standardises numeric values using learned statistics**
- C. Deletes test rows
- D. Predicts labels

A common method to encode nominal data such as city is:

- **A. One-hot encoding**
- B. Median imputation
- C. Train/test split
- D. Accuracy scoring

OneHotEncoder(handle_unknown='ignore') helps when:

- A. All values are missing
- **B. A category appears during transform that was not seen during fitting**
- C. There are no features
- D. A model overfits

Data leakage occurs when:

- A. A notebook is not saved
- **B. Held-out or unavailable information improperly influences training/evaluation**
- C. A dataset has many columns
- D. Accuracy is low

Imputers and scalers should generally be fitted:

- A. On all rows before splitting
- **B. On the test set only**
- C. On the training set only
- D. On unrelated data

In scikit-learn, fit usually means:

- A. Learn parameters or rules from data

- B. Display a chart
- **C. Remove the target automatically**
- D. Save a notebook

In scikit-learn, transform usually means:

- **A. Apply a learned transformation**
- B. Learn from test labels
- C. Create business requirements
- D. Split rows randomly

A benefit of a scikit-learn Pipeline is that it:

- **A. Guarantees 100% accuracy**
- B. Combines steps and applies them consistently
- C. Removes evaluation needs
- D. Makes all missing values harmless

Which problem may be better solved with a simple rule?

- **A. Flag an invoice number that duplicates one for the same vendor**
- B. Predict house price from many historical examples
- C. Discover customer groups from behaviour
- D. Forecast sales from trends

Training and evaluating on the same rows can mislead because:

- A. Labels are always removed
- **B. It may measure fit to seen data rather than generalisation**
- C. It always converts classification to regression
- D. It automatically creates more data

Which statement about scaling is most accurate?

- A. Every model always requires it
- B. No model can use it
- **C**. **It helps many models but is not required for every model**
- D. It replaces test evaluation

Part B — Practical Preprocessing Task (40 marks)

Use the customer churn dataset from the Week 4 Day 2 assignment or the dataset supplied by your trainer. Create Week4_Assessment_Preprocessing_YourName.ipynb.

1. Load the data with pandas. Display the first five rows, shape, data types, missing-value counts, duplicate count, and target distribution. (5 marks)
2. Identify the target and input features. Explain why the target must not be included in X. (5 marks)
3. Split into train and test sets before fitting any imputer, scaler, or encoder. Use a fixed random_state and explain the purpose. (5 marks)
4. Build a numeric pipeline with median imputation and scaling. (7 marks)
5. Build a categorical pipeline with missing-value imputation and OneHotEncoder(handle_unknown='ignore'). (7 marks)
6. Combine the steps using ColumnTransformer. Fit on training data only and transform both train and test data. (6 marks)
7. Print transformed shapes and explain what changed. Identify one possible leakage risk. (5 marks)

code:

```
# 1. Load the data with pandas. Display the first five rows, shape, data types, missing-value counts, duplicate count, and target distribution.

import pandas as pd


df = pd.DataFrame({
    "age": [24, 35, 29, 42, 31, None, 27, 39, 33, 46, 28, 36],
    "monthly_spend": [500, 1200, None, 2200, 900, None,
                      750, 1800, 1100, 2500, 650, 1500],
    "city": ["Hyderabad", "Chennai", "Hyderabad", "Bengaluru",
             "Chennai", "Hyderabad", "Visakhapatnam", "Bengaluru",
             "Chennai", "Hyderabad", "Bengaluru", "Chennai"],
    "plan_type": ["Basic", "Premium", "Standard", "Premium",
                  "Basic", "Basic", "Standard", "Premium",
                  "Standard", "Premium", "Basic", "Standard"],
    "support_tickets": [2, 0, 1, 3, None, 4, 1, 2, 0, 5, 1, 2],
    "churn": ["Yes", "No", "No", "Yes", "No", "Yes",
```

```
        "No", "No", "No", "Yes", "No", "Yes"]
})

df.head()

df.shape

df.dtypes

df.isnull.sum()

df.duplicated().sum()

df["churn"].value_counts()

# 2.Identify the target and input features. Explain why the target must not be included in X.

X = df[[
    "age",
    "monthly_spend",
    "city",
    "plan_type",
    "support_tickets"
]]

y = df["churn"]

# 3. Split into train and test sets before fitting any imputer, scaler, or encoder. Use a fixed random_state and explain the purpose. (5 marks)/

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)
```

4. Build a numeric pipeline with median imputation and scaling. (7 marks)

```
from sklearn.pipeline import Pipeline

from sklearn.impute import SimpleImputer

from sklearn.preprocessing import StandardScaler


numeric_features = [

    "age",

    "monthly_spend",

    "support_tickets"

]


numeric_pipeline = Pipeline(steps=[

    ("imputer", SimpleImputer(strategy="median")),

    ("scaler", StandardScaler())

])
```

5. Build a categorical pipeline with missing-value imputation and OneHotEncoder(handle_unknown='ignore'). (7 marks)

```
from sklearn.preprocessing import OneHotEncoder


categorical_features = [

    "city",

    "plan_type"

]


categorical_pipeline = Pipeline(steps=[

    ("imputer", SimpleImputer(strategy="most_frequent")),

    ("encoder", OneHotEncoder(handle_unknown="ignore"))

])
```

6. Combine the steps using ColumnTransformer. Fit on training data only and transform both train and test data. (6 marks)

```
from sklearn.compose import ColumnTransformer


preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_pipeline, numeric_features),
    ("cat", categorical_pipeline, categorical_features)
])

# 7. Print transformed shapes and explain what changed. Identify one possible leakage risk. (5 marks)
```

Part C — Debugging (15 marks)

1. Scaling before splitting: A student scales the full dataset before splitting into train and test.

Your task: Explain the risk and give the correct sequence. (3 marks)

Scaling the full dataset before splitting can cause data leakage because information from the test data is used. The correct sequence is to split the data first, fit the scaler on training data, and transform both training and test data.

2. Target leakage: A student includes churn in X and also uses it as y.

Your task: Explain why this invalidates the exercise and correct it. (3 marks)

Including churn in both X and y causes target leakage because the model uses the answer itself to make predictions. Keep churn only in y and remove it from X.

3. Unknown category: A test row contains a city not seen during training and the encoder raises an error.

Your task: Name the encoder setting that can help and explain it. (3 marks)

Use OneHotEncoder(handle_unknown='ignore'). It allows unseen categories in test data without raising an error by encoding them as all zeros for that feature.

4. Wrong imputation: A student fills a numeric column with the text 'unknown' and then scales it.

Your task: Explain the type mismatch and suggest a better strategy. (3 marks)

Filling a numeric column with the text 'unknown' causes a type mismatch because scaling requires numeric values. Use SimpleImputer(strategy='mean') or SimpleImputer(strategy='median') to fill missing values before scaling.

5. Misleading evaluation: A student uses training accuracy as proof the model works on new customers.

Your task: Explain why a held-out test set is needed. (3 marks)

Training accuracy shows performance on data the model has already seen. A held-out test set is needed to check how well the model predicts outcomes for new, unseen customers.

Part D — Explain in Your Own Words (10 marks)

1. Explain supervised vs. unsupervised learning, with one example. (2 marks)
   Ans:Supervised Learning: The model learns from labelled data (input and correct output). Example: Predicting house prices using past house data.
   Unsupervised Learning: The model learns from unlabelled data and finds patterns. Example: Grouping customers based on their shopping habits.

2. Explain features vs. labels using one example. (2 marks)
   Ans:Features: The input variables used to make predictions.
   Labels: The output or target that the model predicts.
   Example: In house price prediction, house size and location are features, while the house price is the label

3. Explain why preprocessing should be fitted on training data only. (2 marks)
   Ans:Preprocessing should be fitted only on training data to prevent data leakage. This ensures that information from the test data does not influence the model and helps us evaluate its performance fairly.

4. Explain fit vs. transform. (2 marks)
   Ans;Fit: Learns information from the data, such as the mean used for filling missing values.
   Transform: Uses the learned information to change the data into the required format.
   Example: A scaler learns the mean and standard deviation using training data with fit() and scales the data using transform().

5. Give one problem where a simple rule is preferable to ML. (2 marks)

Ans:ATM PIN verification.

A simple rule is better because the system only needs to check whether the entered PIN matches the stored PIN. ML is unnecessary because this task has a clear, fixed condition.

Part E — Concept Viva (15 marks)

The trainer will ask you to explain your own notebook. Do not simply read your notes.

- Explain your workflow from loading data to transforming it.

Ans:My workflow starts with loading the dataset using Pandas. I check the data, split it into training and testing sets, handle missing values, scale numeric data, and encode categorical data.

- Point to one numeric feature and explain its imputation and scaling.

Ans:For numeric columns, I use median imputation to fill missing values. Then I use StandardScaler to bring the values to a similar scale.

- Print to one categorical feature and explain how encoding changes it.

Ans:For categorical columns, I use OneHotEncoder to convert text categories into numeric values that the machine learning model can understand.

- Explain why the target is excluded from X.

Ans:X contains input features, and y contains the target we want to predict. We exclude the target from X to prevent data leakage.

- Explain how your code avoids fitting preprocessing on the test set.

Ans:First, I split the data. Then I fit preprocessing only on the training data. I use the same learned preprocessing to transform the test data.

- Describe one error you encountered or could encounter and how you would debug it.

Ans:One error is KeyError, which happens when a column name is incorrect or missing. I check the column names using df.columns and correct the name.

Viva scoring: concepts (5), explaining code (5), debugging and reasoning (5).

Student Submission Checklist

- Quiz answers completed.
- Notebook runs top to bottom.
- Inspection outputs are visible.
- Train/test split happens before fitting transformations.
- Numeric and categorical pipelines are implemented.
- ColumnTransformer is used correctly.
- Debugging and own-words answers are included.
- I can explain my notebook without reading every line.